Action Plan for Manuscript Revisions: ProtGram-DirectGCN

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1 Part 1: Code & Repository Enhancements

Objective: Address Reviewer 1's comments on usability, reproducibility, and transparency.

1.1 Overhaul README.md

The current README is too minimal and must be expanded into a complete guide for new users.

Action Plan:

1. **Create an Installation Guide:** Provide clear, step-by-step instructions with the exact Conda and Git commands required to set up the environment from scratch.
2. **Write a Configuration Guide:** Explain the key parameters in configuration/config.py, grouping them by function (e.g., Pipeline Control, GCN Parameters, Benchmarking).
3. **Add a "How to Run" Section:** Include example shell commands for common use cases:
   - Running the full pipeline.
   - Running only the integrated test suite.
   - Running a specific evaluation (e.g., only the main PPI prediction).
4. **Document Project Structure:** Briefly explain the purpose of the main directories (source/, configuration/, results/, etc.) to help users navigate the codebase.

1.2 Add Experimental Details to Configuration

Make all hyperparameters used for the manuscript's experiments explicit and easy to locate.

Action Plan:

1. Add a new section to configuration/config.py named _setup_publication_params.
2. In this section, define all key hyperparameters as class attributes. This directly addresses the reviewer's request for reproducibility.

```
def _setup_publication_params(self):
    """Hyperparameters used for the manuscript's published results."""
    self.MANUSCRIPT_GCN_LR = 0.001
    self.MANUSCRIPT_GCN_EPOCHS = 300
    self.MANUSCRIPT_PPI_MLP_BATCH_SIZE = 2048
    self.MANUSCRIPT_RANDOM_SEED = 42
    # ... add all other relevant parameters
```

Listing 1: Example addition to config.py

1.3 Document Data Preprocessing

Clarify the initial data cleaning steps performed on the UniProt FASTA file.

Action Plan:

1. Create a new file named DATA_PREPARATION.md in the repository root.
2. In this file, detail the procedure used to handle duplicate entries or fragments in the original uniprot_sprot.fasta file before its inclusion in the project.

2 Part 2: New Experiments & Ablation Studies

Objective: Address major concerns from Reviewers 2 and 3 regarding baselines, model validation, and generalizability.

2.1 Expand Baseline Comparisons

The current comparison against only ProtT5 is insufficient. A broader set of baselines is required.

Action Plan:

1. **Integrate ESM-2 Embeddings:**
   - Add logic to the TransformerEmbedder to download and process embeddings from a state-of-the-art ESM model (e.g., esm2_t33_650M_UR50D).
   - Include these embeddings in the final PPI evaluation pipeline.
2. **Integrate a Structure-Based Baseline:**
   - Acquire AlphaFold2-predicted structures for the proteins in the dataset.
   - Implement a simple baseline using structural features. An effective approach would be to use a pre-trained, structure-based GNN like GearNet or to extract features (e.g., pLDDT scores, contact maps) for a simple classifier.

2.2 Conduct Ablation Studies

Quantify the contribution of the model's key architectural components.

Action Plan:

1. **Directed vs. Undirected Graph:** Run the full PPI evaluation using only the undirected graph component (A_undirected_norm_sparse) to demonstrate the value of directed edges.
2. **Gating Mechanism:** Create a simplified version of the DirectGCNLayer where the learnable gating coefficients are removed and replaced with a simple addition. Compare its performance to the full model.
3. **N-gram Hierarchy:** Run the full pipeline three times, using embeddings generated from:
   - n=1 only.
   - n=1 and n=2 combined.
   - n=1, n=2, and n=3 combined.

   This will demonstrate the performance contribution of higher-order n-grams.

2.3 Test Embedding Generalizability

Demonstrate the utility of the learned embeddings on a task other than PPI prediction.

Action Plan:

1. **Implement GO Function Prediction Task:**
   - Acquire Gene Ontology (GO) annotations for the proteins in the dataset.
   - Set up a new multi-label classification experiment where the goal is to predict GO terms from protein embeddings.
   - Evaluate ProtGram-DirectGCN embeddings against the new baselines (ProtT5, ESM-2) on this task.

2.4 Improve Pooling Strategy

Address the concern that mean-pooling is too simplistic for aggregating n-gram embeddings.

Action Plan:

1. **Implement Attention-Based Pooling:**
   - In source/utils/post.py, create a new pooling function that uses a simple attention mechanism to learn weights for each n-gram's contribution to the final protein embedding.
   - Compare the performance of attention-pooling against mean-pooling on the PPI prediction task.

3 Part 3: Interpretability & Manuscript Revisions

Objective: Address feedback from all reviewers regarding the scientific narrative, clarity, and biological relevance of the work.

3.1 Implement Interpretability Analysis

Make the model less of a "black box" by identifying which features it learns are important.

Action Plan:

1. **Identify Salient N-grams:** After training the final PPI model, implement a method to score the importance of each n-gram. One approach is to use Integrated Gradients or a similar feature attribution method on the trained MLP to find which n-grams contribute most to positive predictions.

2. **Visualize Motifs:** Visualize the most influential n-grams as sequence logos or heatmaps to highlight potentially important biological motifs.

3.2 Rewrite and Expand Manuscript Sections

Incorporate the new results and address specific reviewer comments throughout the text.

Action Plan:

1. **Related Work:** Add a new subsection discussing D-SCRIPT, geometric deep learning on protein structures, and other sequence-based transformers like BioBERT.

2. **Methods:** Create a dedicated "Experimental Details" subsection that includes a table listing all key hyperparameters requested by Reviewer 1.

3. **Discussion:**

   - **Address Performance Gap:** Add a paragraph discussing the performance trade-offs between ProtGram-DirectGCN and large PLMs. Emphasize advantages in computational efficiency, interpretability, and independence from external pre-trained models.
   - **Integrate New Results:** Weave the findings from the new experiments (baselines, ablations, GO prediction) into the Results and Discussion sections to build a stronger argument.
   - **Incorporate New Citations:** Add the papers suggested by Reviewer 2 (PMID: 40386382, 39184075, 36823353) to the Discussion section to contextualize future work.

3.3 Address Minor Manuscript Issues

Correct all minor typographical and clarity issues noted by the reviewers.

Action Plan:

1. Find and fix the "Table ??" placeholder.

2. Add a sentence to the caption of Equation 2 clarifying the operations (e.g., "where $\odot$ denotes element-wise multiplication").

3. Create and insert a new figure in the Methods section that visually illustrates the n-gram graph construction process using a short, example amino acid sequence.

4 Suggested Responses to Reviewers

4.1 Response to Reviewer 1

We thank the reviewer for their positive feedback and constructive suggestions, which have significantly improved the clarity and reproducibility of our work. We have addressed all points as follows:

1. **Code Usability (README):** We agree that the original README.md was insufficient. We have completely overhauled it to provide a comprehensive guide, including detailed sections on installation, environment setup (with a note on WSL for Windows users), configuration of the config.py file, and example commands for running the pipeline.

2. **Related Work:** This was an excellent point. We have expanded the "Related Work" section to include a discussion of important recent methods, including D-SCRIPT, geometric deep learning models, and transformer-based approaches like BioBERT, to better contextualize our contribution.

3. **Experimental Details:** To enhance reproducibility, we have added a new "Experimental Details" subsection within the Methods, which now includes a table listing all key hyperparameters used in our experiments, such as learning rates, batch sizes, random seeds, Cluster-GCN parameters, and PCA settings, as requested.

4.2 Response to Reviewer 2

We are grateful for the reviewer's insightful comments and suggestions, which have helped us strengthen the experimental validation and narrative of our manuscript.

Major Concerns:

1. **Comparative Underperformance vs. PLMs:** We agree this is a critical point. We have expanded the Discussion section to explicitly address the performance trade-offs. We now frame ProtGram-DirectGCN not as a direct replacement for large PLMs but as a valuable alternative that is more computationally efficient, does not require massive pre-training, and offers a higher degree of interpretability. We have also broadened our comparative analysis to include ESM-2, providing a more complete picture.

2. **Pooling Strategy:** This is an excellent suggestion. We have implemented an attention-based pooling mechanism as an alternative to mean-pooling and have included the comparative results in the revised manuscript.

3. **Interpretability:** We agree that demonstrating the biological relevance of the learned representations is crucial. We have performed a new interpretability analysis to identify salient n-gram motifs that are most predictive of interactions and have included these findings, along with visualizations, in the Results section.

4. **Generalizability:** To demonstrate the broader utility of our learned embeddings, we have added a new downstream evaluation task: Gene Ontology (GO) function prediction. The results, which show competitive performance, are now included in the manuscript.

5. **New Citations:** We thank the reviewer for these valuable references. We have incorporated the suggested papers (PMID: 40386382, 39184075, 36823353) into our Discussion section to inform future directions.

Minor Concerns:

- We have corrected the "Table ??" placeholder.
- We have added a clarification to the caption of Equation 2 to specify the nature of the operations.
- We have added a new figure to the Methods section that visually illustrates the n-gram graph construction process, as suggested.

4.3 Response to Reviewer 3

We thank the reviewer for their valuable feedback and for recognizing the promise of our framework. We have performed several new experiments to address the points raised.

1. **Ablation Studies:** This is a crucial point that we have now addressed thoroughly. We have conducted a series of ablation studies to quantify the contribution of our model's key components. The new Results section includes analyses comparing: (i) our directed graph versus an undirected version, (ii) our model with and without the gating mechanism, and (iii) performance across different n-gram levels (n=1, n=1,2, and n=1,2,3). These results confirm the importance of each component to the model's final performance.

2. **Generalizability and Use Case:** We agree that demonstrating the utility of our embeddings beyond a single task is important. We have added a new evaluation on a Gene Ontology (GO) function prediction task. The strong performance on this task helps to better define the best use cases for our method and demonstrates the general applicability of the learned representations.

3. **Comparative Analysis:** We have expanded our comparative analysis to provide a more robust evaluation. In addition to ProtT5, we now include results from another state-of-the-art protein language model (ESM-2) and a structure-based baseline method. This broader comparison provides better context for the strengths and weaknesses of our approach.

   "A notable class of modern PPI prediction methods leverages 3D structural information, either from experimental sources or high-fidelity predictions from models like AlphaFold2 [Jumper et al., 2021]. These geometric deep learning approaches, such as GearNet and GVP-GNN, have demonstrated state-of-the-art performance by directly encoding the physical and chemical properties of protein surfaces.While these methods are powerful, their applicability is contingent on the availability of accurate structural data. Our work, with ProtGram-DirectGCN, intentionally explores a different and complementary direction. We focus exclusively on the protein's primary sequence, aiming to develop a method that is (1) universally applicable to any protein, including those with unknown or poorly predicted structures (e.g., intrinsically disordered proteins), and (2) computationally less intensive, as it does not require the computationally expensive step of structure prediction or the storage of large structural files. By constructing a global n-gram graph, our approach seeks to infer higher-order sequence motifs that serve as a proxy for structural and functional information, providing a robust and scalable alternative for large-scale proteome analysis where structural information may be sparse or unavailable. Future work could explore hybrid models that fuse our learned n-gram representations with structural features for proteins where both are available."